

```
In [1]: import pandas as pd
        from sklearn.model_selection import train_test_split
        from sklearn import tree
        from sklearn.tree import DecisionTreeClassifier
        from sklearn import metrics
        import seaborn as sn
        import matplotlib.pyplot as plt
```

```
In [39]: import pandas as pd

        data = {
            'Age': [22, 25, 28, 29, 31, 32, 35, 38, 40, 42, 45, 50],
            'Income': [25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80],
            'Buy_House': [0, 0, 0, 0, 1, 1, 1, 1, 1, 0, 1, 1]
        }

        df = pd.DataFrame(data)

        # Create CSV file
        df.to_csv("house_purchase.csv", index=False)

        print("Dataset created successfully!")
        print("Shape:", df.shape)

        df.head()
```

Dataset created successfully!

Shape: (12, 3)

```
Out[39]:
```

	Age	Income	Buy_House
0	22	25	0
1	25	30	0
2	28	35	0
3	29	40	0
4	31	45	1

```
In [43]: import pandas as pd

        data = pd.read_csv("house_purchase.csv")

        print(data.head())
        print(data.columns)
```

```

    Age  Income  Buy_House
0    22     25         0
1    25     30         0
2    28     35         0
3    29     40         0
4    31     45         1
Index(['Age', 'Income', 'Buy_House'], dtype='object')
```

```
In [45]: X = data[['Age', 'Income']]
        y = data['Buy_House']
```

```
print("X:")
print(X)

print("\nY:")
print(y)
```

X:

	Age	Income
0	22	25
1	25	30
2	28	35
3	29	40
4	31	45
5	32	50
6	35	55
7	38	60
8	40	65
9	42	70
10	45	75
11	50	80

Y:

0	0
1	0
2	0
3	0
4	1
5	1
6	1
7	1
8	1
9	0
10	1
11	1

Name: Buy_House, dtype: int64

```
In [51]: from sklearn.tree import DecisionTreeClassifier
```

```
model = DecisionTreeClassifier(
    criterion='entropy',
    random_state=5
)

model.fit(X, y)

print("Decision Tree trained successfully!")
```

Decision Tree trained successfully!

```
In [53]: y_pred = model.predict(X)
```

```
print("Predicted values:")
print(y_pred)
```

Predicted values:

[0 0 0 0 1 1 1 1 1 0 1 1]

```
In [55]: from sklearn.metrics import accuracy_score
```

```
accuracy = accuracy_score(y, y_pred)
```

```
print("Accuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100)
```

Accuracy: 1.0

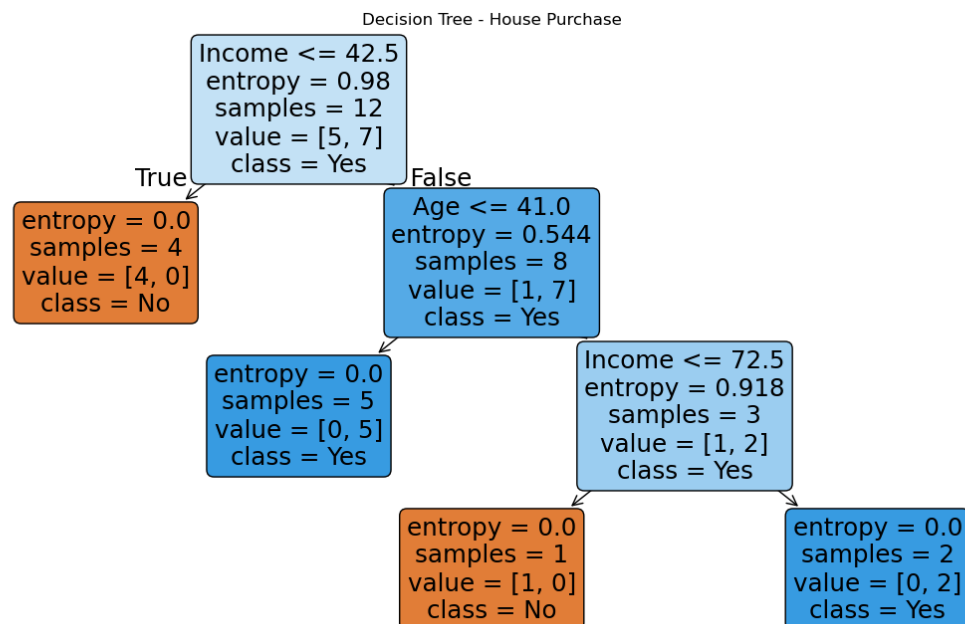
Accuracy Percentage: 100.0

```
In [57]: from sklearn import tree
import matplotlib.pyplot as plt

plt.figure(figsize=(15, 8))

tree.plot_tree(
    model,
    feature_names=['Age', 'Income'],
    class_names=['No', 'Yes'],
    filled=True,
    rounded=True
)

plt.title("Decision Tree - House Purchase")
plt.show()
```



```
In [61]: from sklearn import metrics

# Confusion Matrix
conf_mat = metrics.confusion_matrix(y, y_pred)

print("Confusion Matrix :")
print(conf_mat)
```

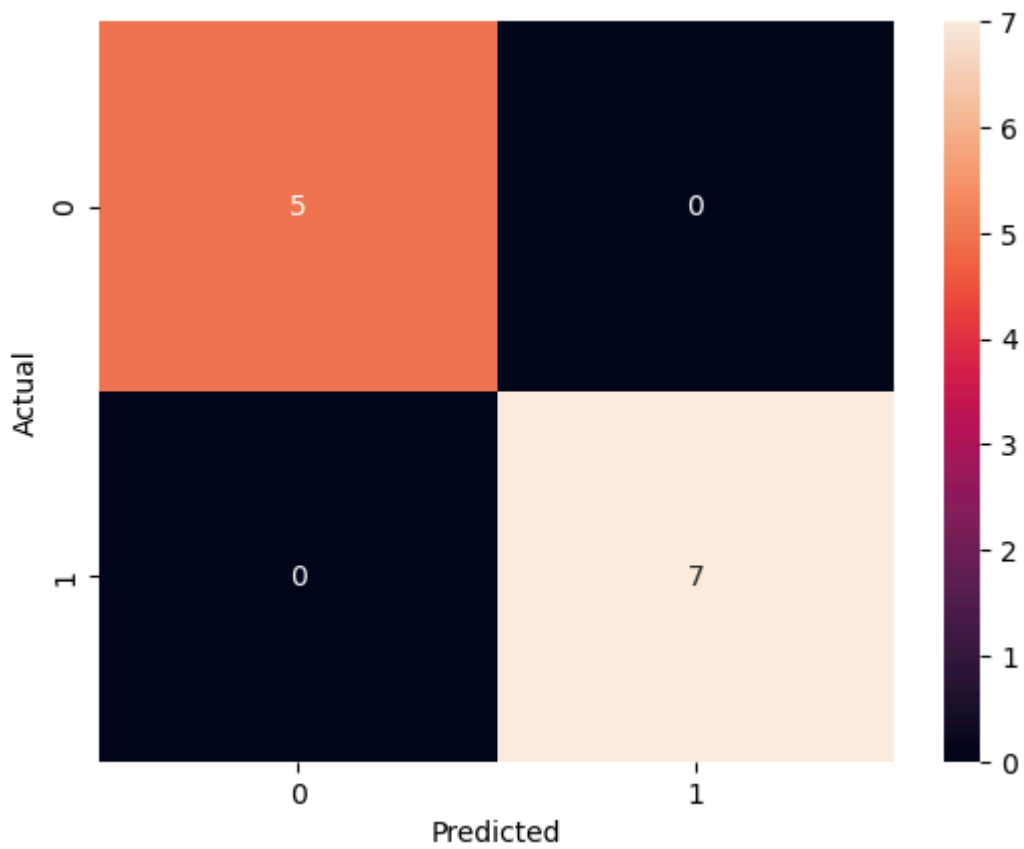
Confusion Matrix :

```
[[5 0]
 [0 7]]
```

```
In [63]: import seaborn as sn
import matplotlib.pyplot as plt

conf_mat = pd.crosstab(
    y,
    y_pred,
    rownames=['Actual'],
```

```
colnames=['Predicted']  
)  
sn.heatmap(conf_mat, annot=True)  
plt.show()
```



In []: